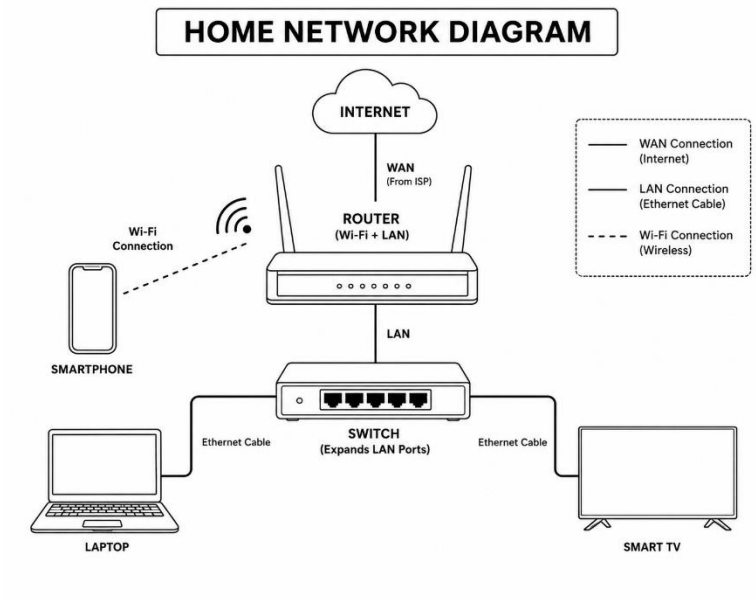
Network devices:-

Tasks:-

1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

Ans.:-



Device	Connection Type	Connected To
Router	Internet (WAN)	ISP/Internet
Switch	Ethernet (LAN)	Router
Laptop	Ethernet	Switch
Smartphone	Wi-Fi	Router
Smart TV	Ethernet	Switch

Working:-

The router receives the internet connection from the ISP. It provides Wi-Fi for wireless devices (smartphone) and a LAN connection to the switch. The switch allows multiple wired devices, such as the laptop and smart TV, to communicate with each other and access the internet through the route

2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Ans.:-

Network Device	Main Function	Connection Type	Real-World Use Case	Daily App/Service Example
Router	Connects the local network (LAN) to the Internet and assigns IP addresses.	Wired & Wireless	Provides Internet access to all devices at home.	YouTube, WhatsApp, Google Chrome
Switch	Connects multiple devices in the same LAN and forwards data only to the intended device.	Wired (Ethernet)	Connects desktop PCs in a computer lab or gaming café for fast communication.	Steam, Valorant, LAN File Sharing
Hub	Connects devices but sends data to every connected device.	Wired (Ethernet)	Used in networking labs to demonstrate how data broadcasting works.	File Sharing, Basic Network Testing
Access Point (AP)	Extends or creates a Wi-Fi network.	Wireless	Increases Wi-Fi coverage in schools, offices, hotels, or large homes.	Netflix, Zoom, Instagram

Modem	Connects the home router to the ISP and converts Internet signals.	Wired	Receives Internet from the ISP and passes it to the router.	All Internet services (YouTube, Gmail, Facebook)
Repeater	Boosts and extends weak Wi-Fi signals.	Wireless	Improves Wi-Fi signal in bedrooms or upper floors.	YouTube, Spotify, WhatsApp Calls
Bridge	Connects two separate LAN segments into one network.	Wired/Wireless	Connects two office buildings or departments on the same network.	Shared Printer, Shared Files
Gateway	Connects networks using different communication protocols.	Wired/Wireless	Allows communication between different types of networks in organizations.	Cloud Services, Microsoft Teams
Network Interface Card (NIC)	Enables a device to connect to a network.	Wired or Wireless	Lets a laptop or desktop access the Internet or LAN.	Google Meet, Chrome, Online Games

3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Ans.:-

Scenario: Ordering Food on Zomato:-

I open the Zomato app on my smartphone to order food from a nearby restaurant. My phone first connects to the Access Point (AP) or the Wi-Fi feature of my home router. The access point receives the wireless data from my phone and forwards it to the local network.

Next, the data reaches the Switch, which transfers the data efficiently between devices on the local network and forwards it to the Router.

The Router assigns my phone an IP address (if needed) and sends the request through the modem to my Internet Service Provider (ISP). The request then travels across the Internet to Zomato's servers, where the restaurant list, menu, and order details are processed.

After I place my order, Zomato's server sends a confirmation back through the Internet to my Router, then to the Switch, then to the Access Point, and finally wirelessly to my smartphone, where I see the order confirmation and delivery tracking.

Role of Each Network Device:-

- Access Point (AP): Connects the smartphone to the Wi-Fi network.
- Switch: Forwards data between devices within the local network.
- Router: Connects the home network to the Internet and routes data to and from Zomato's servers.

**4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup.

Hint: Think about devices like switches, powerline adapters, or WiFi extenders.**

Ans.:-

Solution 1: Wi-Fi Range Extender (with Ethernet Port):-

A Wi-Fi Range Extender receives the Wi-Fi signal from the router and extends it to another area of the house. If the extender has an Ethernet (LAN) port, you can connect your gaming PC to it using an Ethernet cable.

How it works:

1. The router sends a Wi-Fi signal.
2. The Wi-Fi extender receives and boosts the signal.
3. The gaming PC connects to the extender using an Ethernet cable.
4. The PC accesses the Internet through the extender.

Best for**:** Homes where the Wi-Fi signal reaches the PC area but is weak.

Advantages

- Easy to install
- Extends Wi-Fi coverage
- No additional network cables required

Example

Watching YouTube, attending Zoom meetings, browsing websites, and casual online gaming.

Solution 2: Powerline Adapter:-

A Powerline Adapter uses your home's electrical wiring to carry network data.

How it works:

1. Connect the first powerline adapter to the router using an Ethernet cable and plug it into a wall socket.

2. Plug the second adapter into a wall socket near the gaming PC.
3. Connect the second adapter to the PC using an Ethernet cable.
4. The Internet travels through the home's electrical wiring, giving the PC a wired connection.

Best for: Rooms that are far from the router where running a long Ethernet cable is not practical.

Advantages

- Stable wired connection
- Low latency for online gaming
- No need to install long Ethernet cables

Example

Playing Valorant, Counter-Strike 2, PUBG, or downloading games from Steam with a stable connection.